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# Los Angeles River Metals TMDL Implementation Plan for the Cities of Glendale and Burbank:

# Enhanced Street Sweeping Pilot Program Report

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*prepared for:*

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# 1 Introduction

This document constitutes the technical report for the enhanced Street Sweeping Pilot Program (Pilot Program) to be implemented by the City of Burbank and the City Glendale (Cities) as part of the approved *Implementation Plan for the Los Angeles River and Tributaries Total Maximum Daily Load (TMDL) for Metals* (LWA, 2010).

## 1.1 IMPLEMENTATION PLAN BACKGROUND

The California Regional Water Quality Control Board, Los Angeles Region (Regional Board) adopted the Los Angeles River and Tributaries Metals Total Maximum Daily Load (TMDL), Resolution No. 2007-14, on September 6, 2007. Per the requirements of the TMDL, the Cities submitted a draft Implementation Plan to the Regional Board on January 11, 2010. The Implementation Plan outlines the process and implementation actions by which the Cities will jointly achieve the dry weather waste load allocations (WLAs) for copper and lead and the wet weather WLAs for copper, lead, zinc, and cadmium within their jurisdictions. The draft Implementation Plan was approved, with recommended changes, by the Regional Board in a letter dated June 14, 2010. A final Implementation Plan, *Implementation Plan for the Los Angeles River and Tributaries Total Maximum Daily Load (TMDL) for Metals* (LWA, 2010), was submitted by the Cities to the Regional Board on August 2, 2010.

The Implementation Plan is being conducted through a phased approach (**Table 1**) based on optimizing implementation efforts by iteratively implementing three tiered elements:

- Regulatory Approaches (Special Studies)
- Source Controls (Point and Non-Point Sources)
- Structural Controls (BMPs/Treatment Facilities)

**Table 1. City of Burbank and City of Glendale Los Angeles River Metals TMDL Implementation Plan Phases, Time Period, and Goals**

| Phase | Implementation Time Period | Goals |
|---|---|---|
| Phase I | Present – January 11, 2012 | 50% of area meets dry weather WLA<br>25% of area meets wet weather WLA |
| | January 11, 2012 – January 11, 2020 | 75% of area meets dry weather WLA |
| Phase II | January 12, 2020 – January 11, 2024 | 100% of area meets dry weather WLA<br>50% of area meets wet weather WLA |
| Phase III | January 12, 2024 – January 11, 2028 | 100% of area meets both dry and wet weather WLAs |

For Phase I, the Cities committed to certain implementation actions, including a Pilot Program to evaluate potential enhancements to the existing street sweeping programs for each of the Cities (see the Implementation Plan, Section 4.1.2.3). The Pilot Program is intended to evaluate the effectiveness of the existing sweeping programs and to inform management decisions regarding the appropriateness and benefit of implementing enhancements to the sweeping programs.

The Cities commenced development of the Pilot Program in August 2010. Sample collection, which is further described in Section 3.2, occurred from February 2011 through September 2011. As the sampling events focused on dry-weather days, the evaluation period was extended through September to accommodate the multiple rain events that occurred during the spring of 2011.

## 1.2 TECHNICAL BACKGROUND

Mechanized street sweeping is typically used to remove debris (i.e., trash and sediment) from roadways and other paved surfaces. However, heavy metals and other pollutants (i.e., nutrients and organics), are attached to street sediment, referred to as "street dirt." Studies dating back to the 1970's (Sartor and Boyd, 1972; Pitt and Amy, 1973; Pitt, 1979; Revitt and Ellis, 1980; Biggins and Harrison, 1980; Pitt and Sutherland, 1982; Pitt and Shawley, 1982; Pitt, 1985; Pitt and McLean, 1986; HDR et. al., 1993; Sansalone, et. al., 1998; Sansalone and Tribouillard, 1999; Hubell, Roth and Clark, 2001; Tetra Tech, 2001; Townsend, et. al., 2002; and PWR, 2004) have consistently demonstrated that up to 50 percent of heavy metals and other pollutants are attached to "street dirt" whose particles are too small for common sweeping practices to remove effectively.

Various studies (Tetra Tech and Pacific Water Resources 2001, SPU 2009) have demonstrated that enhancements to sweeping programs can result in greater pollutant removal efficiencies and reductions in metals loadings as well as loading from other pollutants (i.e., nutrients and organics). Such enhancements can include using sweepers with increased pickup efficiencies, increased sweeping frequency, posting of streets for sweeping to remove cars that may impede street sweepers as well as increased enforcement, and modification to sweeping methods (i.e., altered driving speed). The following factors have the potential to affect the overall performance of a street sweeping program and evaluating and addressing these factors may allow for effective enhancement of a program:

1. **Sweeper Type:** There are four general types of sweepers: mechanical broom, vacuum, regenerative air, and high efficiency; all of which have varying degrees of pickup efficiencies. Key effects of sweeper type on performance include the particle size collected and fugitive dust loss, which occurs when street dirt is lost during the sweeping process.
2. **Forward Speed:** The forward speed of a street sweeper will significantly affect its ability to pick up particulate material.
3. **Sweeping Frequency:** The frequency of sweeping can affect the amount of street dirt that accumulates on street surfaces and has the potential to runoff into receiving waters.
4. **Parked Car Interference:** As the majority of street dirt accumulates within a few feet of the curb (USGS 2007; Pitt, 1979; Sartor and Boyd, 1972), most street dirt accumulates within the width of a parked car. Therefore, access to the curb is essential for effective street sweeping.
5. **Barriers:** As the majority of street dirt accumulates within a few feet of the curb (USGS 2007; Pitt, 1979; Sartor and Boyd, 1972), street sweeping is the most effective on streets that are curbed.

6. **Pavement Conditions:** Uneven surfaces that accompany poor pavement conditions make it difficult for sweepers to operate effectively, especially the regenerative air sweeper, as dirt can accumulate in surface cracks and deep depressions.

# 2 Pilot Program Approach and Methodology

The goal of the Pilot Program was to identify methods by which to enhance the existing sweeping programs within the Cities to attain or support the attainment of the WLA requirements of the TMDL while minimizing the implementation of costly structural control measures. Given the ability of the Cities' sweeping programs to affect the six factors discussed in the previous section, the Pilot Program focused on evaluating potential enhancements based on the first four factors over which the sweeping programs have the most control: sweeper type, forward speed, sweeping frequency, and parked car interference.

To assess potential enhancements, the Pilot Program was conducted over three phases:

- Phase I: Establish a baseline for the effectiveness of the current sweeping program in the removal of street dirt and attached metals
- Phase II: Evaluate potential enhancements to the existing program in a controlled environment
- Phase III: Evaluate potential enhancements to the existing program in the field

The results of each Phase informed the approach for the subsequent Phases. Therefore, this report presents the approach and methodology, results, and discussion by each Phase of the Pilot Program.

# 3 Phase I: Baseline of Current Sweeping Program

Phase I focused on collecting street dirt at various land uses within each City and analyzing the samples for total loading, physical characteristics (particle size distribution), and chemical characteristics (metals concentrations) to establish a baseline of the effectiveness of the Cities' current street sweeping programs. Street dirt samples were collected before the regularly scheduled sweeping day (referred to as "pre-sweeping" samples) and after the street was swept (referred to as "post-sweeping" samples). Samples were collected randomly, though pre- and post-sweeping samples were collected from different areas within each site to allow comparison of effectiveness. Under Phase I, sweeper operators were not informed that the Pilot Program was underway to avoid potential bias.

## 3.1 APPROACH

Ten sites with various land use types were evaluated, with five sites evaluated in each City (**Table 2**). Four of the sites were characterized as residential, two as commercial, two as industrial, and two as major arterial. Although other land uses exist within the Cities (**Table 3**), the ten sites selected are representative of over 90 percent of the various combinations of land use and street characteristics covered by the Cities' sweeping programs. **Table 4** presents the characteristics of the ten sites and **Figure 1** depicts the locations of the sites as well as the corresponding land use areas sampled.

**Table 2. Phase I Sampling Sites and Land Use Types**

| City | Site Name | Site ID | Land Use |
|---|---|---|---|
| Burbank | Jeffries Ave | JEFF | Single Family Residential (SFR) |
| | Clark Ave | CLARK | Single Family Residential (SFR) |
| | W Magnolia Blvd | MAG | Commercial (COM) |
| | Winona Ave | WIN | Industrial (IND) |
| | N Hollywood Way | HOLL | Major Arterial (ART) |
| Glendale | W Kenneth Rd | KENN | Single Family Residential (SFR) |
| | Kirkham Dr | KIRK | Single Family Residential (SFR) |
| | N Orange St | ORNG | Commercial (COM) |
| | Grand Central Ave | GRAND | Industrial (IND) |
| | N Verdugo Rd | VERD | Major Arterial (ART) |

**Table 3. Land Use Types for the Cities of Burbank and Glendale**

| City | Land Use Type | Area (Acres) | Percent of Total |
|---|---|---|---|
| Burbank | Residential | 5,089 | 45.9% |
| | Commercial | 1,188 | 10.7% |
| | Industrial | 655 | 5.9% |
| | Major Arterial | 323 | 2.9% |
| | Open Space/ Recreation | 2,846 | 25.7% |
| | Other Land Uses | 1,304 | 8.9% |
| Glendale | Residential | 8,106 | 41.4% |
| | Commercial | 1,292 | 6.6% |
| | Industrial | 997 | 5.1% |
| | Major Arterial | 260 | 1.3% |
| | Open Space/ Recreation | 8,267 | 42.3% |
| | Other Land Uses | 643 | 3.3% |

**Table 4. Characteristics of Phase I Sampling Sites**

| City | Site ID | Land Use | Area (acres) | Curb Length (miles) | Curb Density (ft/acre) | Sweeping Frequency | Sweeper Type Used[1] | Posted No Parking? | Parking Restrictions Enforced? |
|---|---|---|---|---|---|---|---|---|---|
| Burbank | JEFF | SFR | 54 | 1.4 | 135 | Weekly | Mech | Y | Y |
| | CLARK | SFR | 78 | 2.0 | 137 | Weekly | Mech | Y | Y |
| | WIN | IND | 120 | 1.3 | 58 | Weekly | Mech | Y | Y |
| | MAG | COM | 9.7 | 1.0 | 544 | Weekly | Mech | Y | Y |
| | HOLL | ART | 10 | 1.0 | 528 | Weekly | Mech | Y | Y |
| Glendale | KIRK | SFR | 37 | 1.3 | 188 | Every 2 Weeks | Mech | N | NA |
| | KENN | SFR | 61 | 1.6 | 138 | Every 2 Weeks | Mech | N | NA |
| | GRAND | IND | 89 | 1.5 | 89 | Every 2 Weeks | Mech | N | NA |
| | ORNGE | COM | 45 | 1.3 | 157 | 4x Per Week | Mech | N | NA |
| | VERD | ART | 9.1 | 1.0 | 580 | Every 2 Weeks | Mech | N | NA |

1 – The Cities employ several types of sweeper s (mechanical, vacuum, regenerative air), but the predominant sweeper type for both Cities is mechanical so it is assumed that most of the sites are swept using mechanical sweepers.

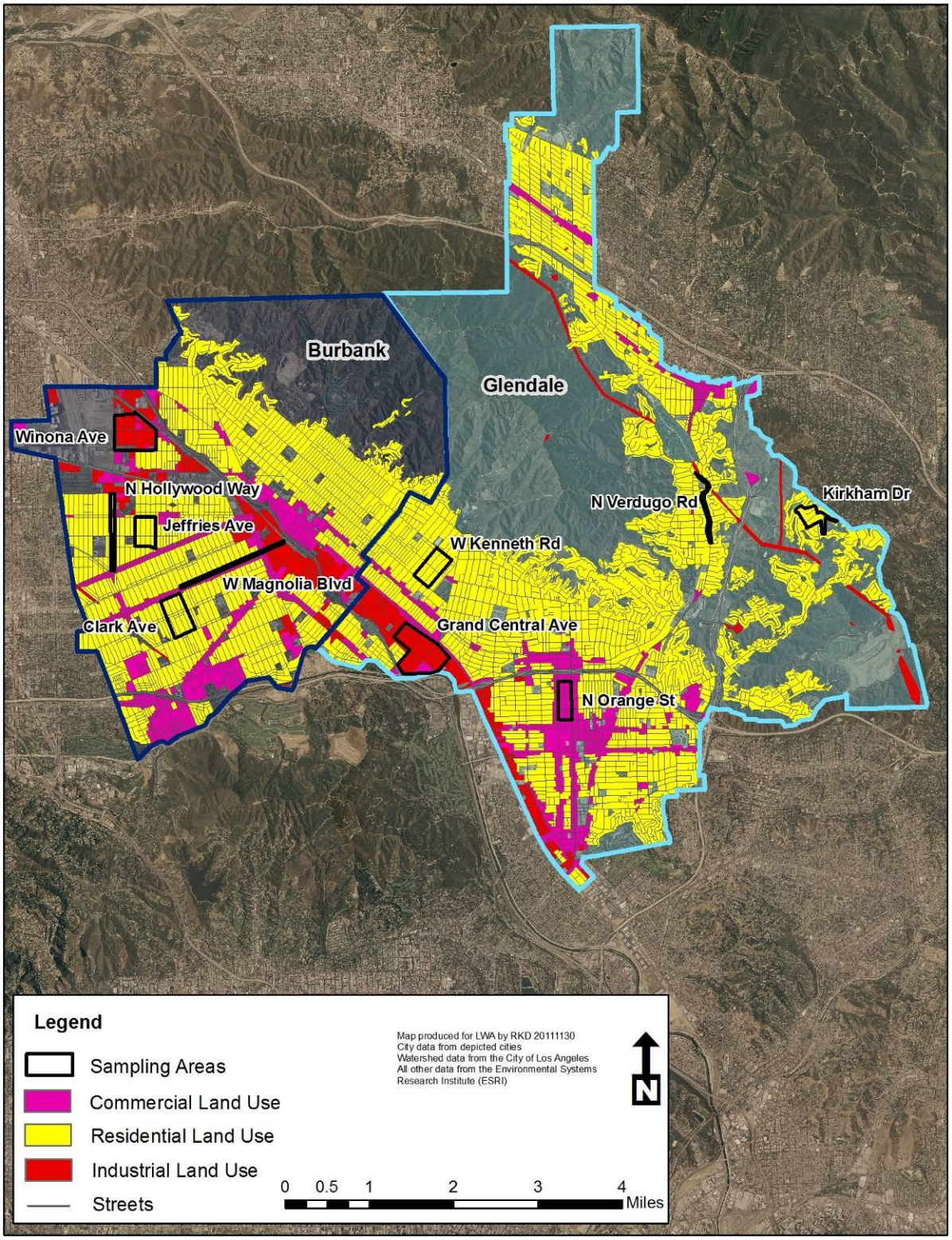


**Figure 1. Sampling Sites, Sampled Land Use Areas, and Streets**

Each site was sampled during three discrete events, with one pre-sweeping and one post-sweeping sample collected per event. Pre-sweeping sampling occurred prior to the regularly scheduled street sweeping and post-sweeping sampling occurred after the area was swept. Pre-sweeping sampling occurred the day before the area was swept at all sites except W Kenneth Rd in Glendale. For W Kenneth Rd, sweeping times were posted, starting at 8am on Mondays. Samples at that site were therefore collected on the same day, prior to 8am. Aside from W Kenneth Rd, most sweeping in the Cities is not scheduled to a predictable hour. As such, the samples were collected one day prior to the scheduled sweeping day to ensure the sweeper operators were not aware of the Pilot Program to avoid potential bias. Post-sweeping samples were collected the day after the area was swept at all sites except Winona Ave which was collected in the afternoon on the same day the site was swept. Pre- and post-sweeping samples were collected only during dry weather conditions. **Table 5** lists the targeted sweeping days, the pre-sweeping sampling days, and post-sweeping sampling days for each site.

**Table 5. Targeted Sweeping Days, Pre-Sweeping Sampling Days, and Post-Sweeping Sampling Days for Each Sampling Site**

| City | Site Name | Pre-Sweeping Sampling Day | Target Sweeping Day[1] | Post-Sweeping Sampling Day |
|---|---|---|---|---|
| Burbank | Jeffries Ave (SFR) | Tuesday | Wednesday | Thursday |
| | Clark Ave (SFR) | Monday | Tuesday | Wednesday |
| | W Magnolia Blvd (COM) | Monday | Tuesday | Wednesday |
| | Winona Ave (IND) | Wednesday | Thursday | Thursday |
| | N Hollywood Way | Tuesday | Wednesday | Thursday |
| Glendale | W Kenneth Rd (SFR) | Monday | Monday | Tuesday |
| | Kirkham Dr (SFR) | Monday | Tuesday | Wednesday |
| | N Orange St (COM) | Tuesday | Wednesday | Thursday |
| | Grand Central Ave (IND) | Monday | Tuesday | Wednesday |
| | N Verdugo Rd (ART) | Tuesday | Wednesday | Thursday |

1 – Several of the areas are swept more than once per week, therefore one sweeping day was targeted to conduct the pre-sweeping sampling and the post-sweeping sampling.

## 3.2 SAMPLE COLLECTION

Each street dirt sample was collected from five 10 feet (ft) by 10 ft areas (500 sq ft total), with each area measuring 10 ft along the curb and 10 ft from the curb into the street. **Figure 2** depicts how each area was demarcated for sample collection. The sampled areas at each site were randomly chosen. Areas sampled during the pre-sweeping sample collection were excluded from the areas sampled for the post-sweeping sample. The sections were clearly marked to prevent sampling the same section during pre-sweeping and post-sweeping sampling.

**Figure 2. Demarcation Method for Phase I Sample Collection**

Samples were collected by vacuuming from the curb to the end of the demarcated section in discrete "pulls" (**Figure 3)** using a stainless steel Shop-Vac® fitted with a Dacron filter bag to keep any fine materials from entering the filter/motor unit**.** Each "pull" was 5.5 inches wide and approximately 22 "pulls" were required to sample one 10 feet by 10 feet area. The collected street dirt was then transferred from the Shop-Vac® canister into a clean Ziploc® bag (**Figure 4**). The samples were labeled with the phase, the Site ID, and a unique sample identification and then stored securely by sealing the Ziploc® bag with tape and placing it in another Ziplock® bag (**Figure 5**). Field logs and the sample collection log were filled out at this time.

**Figure 3. Phase I Sample Collection**

**Figure 4. Sample Transfer Process**

**Figure 5. Sample Storage Procedure**

## 3.3 ANALYTICAL METHODS

All pre-sweeping and post-sweeping samples were analyzed at Physis Environmental Laboratories, Inc. Samples were analyzed for total loading (e.g., the total mass of street dirt collected), particle size distribution (PSD), and metals concentrations.

To determine the PSD, each sample was weighed in the laboratory and then dried in an oven kept below 120° F to avoid adverse chemical transformations. Each sample was then sifted through a series of seven stainless steel sieves with various sizes (**Table 6**). The fractions that passed through each sieve were weighed separately to determine the PSD. PSD was determined for each pre-sweeping and post-sweeping sample individually, for a total of 60 samples [10 sites x 3 events x 2 samples (one pre- one post-sweeping) per event].

**Table 6. Sieve sizes and particle size ranges.**

| Sieve Size | Corresponding Microns |
|---|---|
| ¼ | > 6370 |
| 10 | 2000 – 6370 |
| 18 | 1000 – 2000 |
| 30 | 600 – 1000 |
| 60 | 250 – 600 |
| 120 | 125 – 250 |
| 230 | 63 – 125 |
| Pan | < 63 |

For chemical analyses, the samples were composited by sample type (e.g., pre- or post-sweeping) for each site, yielding one pre-sweeping and one post-sweeping sample per site. Each sample was sieved into three size groups: Fine (less than 63 microns), Medium (63 to 250 microns), and Coarse (250 to 2000 microns). Chemical analyses were then conducted for each size group using EPA analysis method 6020 with a detection limit of 0.025 µg/L and a reporting limit of 0.05 µg/L.

## 3.4 RESULTS

Phase I data (baseline of current sweeping program) were analyzed for pre-sweeping and post-sweeping street dirt loadings, pre-sweeping and post-sweeping PSD, and metals concentrations. The following sections detail the results for these three data categories.

### 3.4.1 Street Dirt Loadings

Average street dirt loadings and percent pick up were calculated for each site (**Table 7**). Average loadings for both pre-sweeping and post-sweeping loadings, in pounds per curb mile, were calculated by converting the mass of street dirt collected from grams to pounds and converting the length of curb where the samples were collected from feet to miles and then dividing the mass in pounds by the curb miles. Average amount removed (percent pick up) was calculated for each site in two ways: (1) using all of the sampling events and (2) using only the sampling events that had positive removal values (e.g., less street dirt in the post-sweeping sample than in the pre-sweeping sample). **Appendix A** presents the data for each pre-sweeping and post-sweeping sample for the City of Burbank and **Appendix B** presents the data for each pre-sweeping and post-sweeping sample for the City of Glendale.

For the City of Burbank, there were five negative removals out of the 15 total pre-sweeping and post-sweeping removal comparisons. Negative removals occur when more street dirt is found in the post-sweeping sample compared to the pre-sweeping sample. A negative removal does not mean that the sweeping program added street dirt to the sample; there are several reasons why negative removals occur as discussed in **Section 3.5**. Despite the negative removals, the average removal for each site was positive and varied from 16 lbs/ curb mile to 68 lbs/ curb mile. In addition, the average percent pick-up using data from all events were positive and ranged from 1.3% to 25%. The average percent pick-up using data from only non-negative events ranged from 11.7% to 25% (**Table 7**).

For the City of Glendale, there were eight negative removals out of the 15 total pre-sweeping and post-sweeping removal comparisons. The average removal for each site varied from -72 lbs/ curb mile to 146 lbs/ curb mile. In addition, the average percent pick-up using data from all events ranged from -57% to 44%. The average percent pick-up using data from only non-negative events ranged from 7% to 44% (**Table 7**).

**Table 7. Average Street Dirt Loadings and Percent Pick-Up by Sampling Site**

| | | | Avg. Loading (lbs/curb mile) | | | Avg. % Pick-Up | | |
|---|---|---|---|---|---|---|---|---|
| City | Site ID | Land Use | Pre-Sweeping | Post-Sweeping | Removal | All Events | Non-Neg. Events | # of Neg. Removals |
| Burbank | CLARK | SFR | 309 | 255 | 54 | 12.7 | 13.3 | 1 of 3 |
| | JEFF | SFR | 267 | 199 | 68 | 25 | 25 | 0 of 3 |
| | MAG | COM | 404 | 375 | 29 | 4 | 15 | 2 of 3 |
| | WIN | IND | 238 | 210 | 28 | 11.7 | 11.7 | 0 of 3 |
| | HOLL | ART | 170 | 154 | 16 | 1.3 | 14 | 2 of 3 |
| Glendale | KENN | SFR | 442 | 434 | 8 | -10 | 24.3 | 2 of 3 |
| | KIRK | SFR | 138 | 210 | -72 | -57 | 7 | 2 of 3 |
| | ORNGE | COM | 83 | 88 | -6 | -15.7 | 15.7 | 2 of 3 |
| | GRAND | IND | 321 | 176 | 146 | 44 | 44 | 0 of 3 |
| | VERD | ART | 340 | 411 | -72 | -25.7 | 6 | 2 of 3 |

### 3.4.2 Physical Characteristics

Pre-sweeping and post-sweeping samples from each site were analyzed to determine the relative weight and percent distribution of the street dirt loadings in eight particle size groups. **Table 8** lists the particle size groups and corresponding size range in microns by which the samples were categorized, with Group 1 containing the finest materials and Group 8 containing the coarsest materials. **Figure 6** presents the average pre-sweeping and post-sweeping PSDs for all sites combined and shows a slight shift to a finer distribution materials in the post-sweeping samples compared to pre-sweeping samples, with the shift occurring around 600 microns. **Table 9** lists the pre-sweeping and post-sweeping PSDs by percent for each site.

**Table 8. Particle Size Groups**

| Group | Size Range (microns) |
|---|---|
| 1 | < 63 |
| 2 | 63-125 |
| 3 | 125-250 |
| 4 | 250-600 |
| 5 | 600-1000 |
| 6 | 1000-2000 |
| 7 | 2000-6300 |
| 8 | > 6300 |

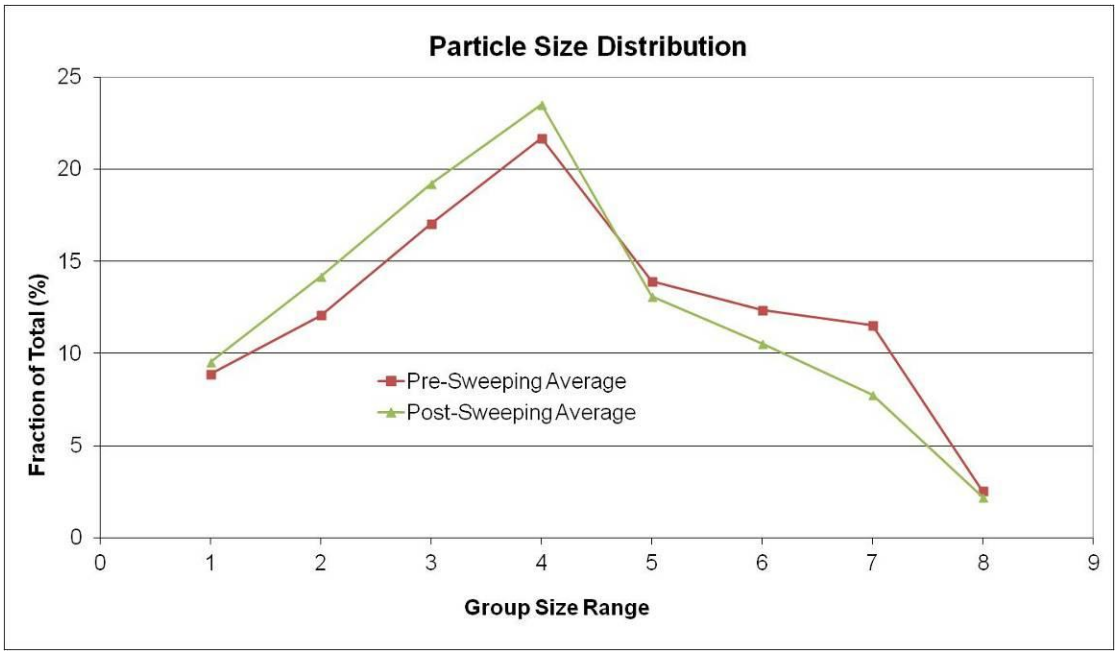


**Figure 6. Average Particle Size Distributions – Pre-Sweeping versus Post-Sweeping**

**Table 9. Average Mass-Fraction Percent by Particle Size Group for Pre-Sweeping and Post-Sweeping Samples by Site**

| | | | Fine | Medium | | | Coarse | | Not Classified | | Percent of Total | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Site Name | Land Use | Sampling Type | <63 (µm) | 63-125 (µm) | 125-250 (µm) | 250-600 (µm) | 600-1000 (µm) | 1000-2000 (µm) | 2000-6300 (µm) | >6300 (µm) | < 63-600 (µm) | > 600 (µm) |
| Burbank Sites | | | | | | | | | | | | |
| CLARK | SFR | PRE | 10.6 | 12.4 | 18.6 | 24.2 | 15.3 | 10.3 | 6.9 | 1.7 | 65.7% | 34.3 % |
| CLARK | SFR | POST | 11.2 | 14.1 | 19.0 | 23.5 | 12.6 | 10.4 | 7.3 | 1.9 | 67.8% | 32.2 % |
| JEFF | SFR | PRE | 12.0 | 15.1 | 18.2 | 18.1 | 10.9 | 9.5 | 9.6 | 6.6 | 63.4% | 36.6 % |
| JEFF | SFR | POST | 13.4 | 19.5 | 21.2 | 20.6 | 9.7 | 7.6 | 5.4 | 2.7 | 74.6% | 25.4 % |
| MAG | COM | PRE | 8.1 | 14.4 | 17.7 | 21.4 | 13.4 | 9.8 | 13.4 | 1.8 | 61.7% | 38.3 % |
| MAG | COM | POST | 9.4 | 17.2 | 21.6 | 25.3 | 14.0 | 7.7 | 3.8 | 0.9 | 73.5% | 26.5 % |
| WIN | IND | PRE | 13.0 | 15.8 | 20.6 | 22.0 | 12.3 | 9.7 | 5.4 | 1.2 | 71.5% | 28.5 % |
| WIN | IND | POST | 13.0 | 16.9 | 19.9 | 20.3 | 11.1 | 9.7 | 6.2 | 2.9 | 70.1% | 29.9 % |
| HOLL | ART | PRE | 8.3 | 13.2 | 20.2 | 25.0 | 13.7 | 10.9 | 7.0 | 1.6 | 66.7% | 33.3 % |
| HOLL | ART | POST | 9.3 | 15.2 | 21.6 | 24.6 | 12.3 | 8.9 | 5.9 | 2.3 | 70.6% | 29.4 % |
| Glendale Sites | | | | | | | | | | | | |
| GRAND | IND | PRE | 15.1 | 15.3 | 15.2 | 15.7 | 10.2 | 10.6 | 13.7 | 4.1 | 61.4% | 38.6 % |
| GRAND | IND | POST | 14.6 | 15.6 | 15.9 | 14.5 | 10.2 | 11.8 | 12.8 | 4.7 | 60.6% | 39.4 % |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| KENN | SFR | PRE | 4.5 | 8.3 | 14.5 | 23.1 | 15.5 | 14.6 | 16.4 | 3.2 | 50.3% | 49.7 % |
| KENN | SFR | POST | 6.3 | 11.8 | 18.7 | 26.0 | 14.2 | 11.7 | 9.8 | 1.6 | 62.8% | 37.2 % |
| KIRK | SFR | PRE | 4.2 | 7.8 | 16.7 | 23.5 | 17.5 | 15.7 | 12.9 | 1.7 | 52.2% | 47.8 % |
| KIRK | SFR | POST | 4.9 | 8.7 | 20.1 | 30.3 | 17.7 | 11.4 | 6.5 | 0.5 | 64.0% | 36.0 % |
| ORNGE | COM | PRE | 8.5 | 10.9 | 14.1 | 17.9 | 13.1 | 16.3 | 17.3 | 2.1 | 51.3% | 48.7 % |
| ORNGE | COM | POST | 7.2 | 12.5 | 16.7 | 21.4 | 13.0 | 13.7 | 12.7 | 2.8 | 57.8% | 42.2 % |
| VERD | ART | PRE | 4.7 | 7.5 | 14.6 | 26.0 | 17.3 | 16.0 | 12.7 | 1.2 | 52.8% | 47.2 % |
| VERD | ART | POST | 6.1 | 10.3 | 17.4 | 28.8 | 16.2 | 12.6 | 7.2 | 1.6 | 62.5% | 37.5 % |

Notes: Grey highlighted cells show the percent of total for the < 63-600 microns mass-fraction and the percent of total for the > 600 microns mass-fraction.

### 3.4.3 Chemical Characteristics

Chemical analyses were conducted on pre-sweeping and post-sweeping samples for cadmium, copper, lead, and zinc (See **Section 3.3**). The average street dirt metal concentrations were determined by land use and particle size group (**Table 10**). Concentrations are based upon the average across the same land use in both Cities [e.g., ART is an average of HOLL (Burbank) and VERD (Glendale)].

For all metals in all of the land uses, the highest average concentrations were found in the Fine particle size group (< 63 microns). Generally, the second highest average concentrations were found in the Medium particle size group (63-250 microns), except for the industrial land use where the second highest concentrations for copper and zinc were in the Coarse particle size group (250-2000 microns).

The highest concentration of zinc was in the COM land use, the highest concentration of lead was in the ART land use, and the highest concentrations of copper and cadmium were in the IND land use (**Table 10**). The SFR land use had the lowest concentration for all metals in all particle sizes. The copper concentrations in the IND land use were not only the highest measured in all the land uses, but the concentrations in all particle sizes were also significantly higher than the other land uses. Cadmium in the IND land use is also significantly higher compared to the other land uses, but only in the fine particle size group.

**Table 10. Average Metal Concentration by Particle Size Group and Land Use. Bold values indicate highest concentration per particle size group.**

| | | Concentration (µg/g) | | | |
|---|---|---|---|---|---|
| **Metal** | **Particle Size Group** | **ART** | **COM** | **IND** | **SFR** |
| Cadmium (Cd) | Fine (< 63 microns) | 1.82 | 1.76 | **6.02** | 1.1 |
| | Medium (63-250 microns) | 0.58 | 1.01 | **2.66** | 0.41 |
| | Coarse (250-2000 microns) | 0.14 | 0.21 | **0.87** | 0.19 |
| Copper (Cu) | Fine (< 63 microns) | 287 | 339 | **1,525** | 120 |
| | Medium (63-250 microns) | 114 | 320 | **1,179** | 70.3 |
| | Coarse (250-2000 microns) | 58.5 | 94.4 | **1,186** | 25.2 |
| Lead (Pb) | Fine (< 63 microns) | **174** | 137 | 86.9 | 81.3 |
| | Medium (63-250 microns) | 87.2 | **96.0** | 56.4 | 42.0 |
| | Coarse (250-2000 microns) | **38.5** | 33.5 | 25.6 | 16.1 |
| Zinc (Zn) | Fine (< 63 microns) | 811 | **947** | 524 | 411 |
| | Medium (63-250 microns) | 315 | **402** | 284 | 182 |
| | Coarse (250-2000 microns) | 115 | 137 | **301** | 97.4 |

## 3.5 DISCUSSION (EFFECTIVENESS OF CURRENT SWEEPING PROGRAMS)

Average street dirt percent pick-up results, PSD results, and metals concentrations results were used in combination to evaluate the overall effectiveness of each City's current sweeping program at removing street dirt and associated metals from the targeted land uses. Average percent pick-up results were used to determine how effective the Cities are at removing street

dirt as a whole while PSD results were used to determine how effective the Cities are at removing specific sized particles. Lastly, metals concentrations results were analyzed to identify the total amount of metals removed.

### 3.5.1 Average Percent Pick-Up

The average percent pick-up differs between the two Cities. The City of Burbank's program had positive average street dirt pick-up results for all of its five sites with a range of 1.3% to 25%. The City of Glendale's average percent pick-up results were more variable with the industrial site (GRAND) achieving the highest average removal effectiveness (44%) and the single family residential site (KIRK) achieving the lowest average removal effectiveness (-57%) of all of the ten sites monitored. Three sites in the City of Glendale averaged an overall negative removal. As stated in **Section 3.4.1**, negative removals occur when more material is observed in post-sweeping samples when compared to pre-sweeping samples. Negative removals can occur for a number of reasons.

First, negative removals have been observed when rain occurs between the pre-sweeping and post-sweeping samplings and sediment and its associated pollutants are transported from adjacent urban surfaces to the street where it is deposited. However, since no rainfall occurred between any of the pre-sweeping and post-sweeping sampling events, sediment transport and deposition is not the cause for the negative removals.

Negative removals can also occur when pre-sweeping street dirt loadings are below the minimum threshold of material needed for the sweeper to achieve any removal. Pitt (1979) showed that there is a minimum amount of street dirt loading below which the sweeper cannot remove additional material. This minimum threshold amount is a function of the type of sweeper and certain operational characteristics. It is also known to vary by particle size group (Sutherland and Jelen, 1997). This seems to be what occurred at the commercial site (ORNGE) in the City of Glendale since the site is swept four times a week. The ORNGE site had the lowest pre-sweeping and post-sweeping loadings of all of the sites monitored and had an average negative removal of -16%.

A third potential reason negative removals were observed is the variability found in street dirt loadings due to the street dirt sampling technique that was employed where samples were randomly collected within each site. If the sampling variability is greater than the average reduction in street dirt due to sweeping, then nothing will appear to have been removed. And even in the case where positive removals are recorded, which occurred in 17 of the 30 monitored sweeping events, sampling variability could be contributing to some of the positive removal that is being assumed.

But if excessive variability was the predominate reason that negative removals were observed, then greater removals would be expected in the City of Glendale (excluding the ORNGE site) than in the City of Burbank. This would be expected since the sweeping frequency is every two weeks in the City of Glendale and weekly in the City of Burbank. The less frequent sweeping schedule in the City of Glendale means there would be more time for the Glendale sites to accumulate street dirt between sweepings, which translates to more material accumulated and more material available for potential removal by sweeping. However, that was not the case since the average removal on the three sites (excluding ORNGE) in the City of Glendale that experienced negative removal sweeping events was -28% compared to Burbank's 6% for its three sites where negative removals were recorded.

Lastly, and likely the predominate issue resulting in observations of negative removals, is that for many of the sites the sweeper is not getting to the curb due to parked car interference. Therefore, during the post-sweeping sampling, significant lengths of curb and adjacent street pavements that did not get swept were sampled.

During several of the post-sweeping sampling events, sweeper trails could be seen especially where the machine maneuvered to avoid parked cars. Due to the random nature of the sampling, it was inappropriate to purposefully select locations where it was known that the sweeper was able to access the curb; therefore, locations that were not swept were included in the sampling events.

Additionally, there was a problem with parked car interference during post-sweeping sampling events in that there were areas that the sampling team could not access, which limited the areas that could be randomly sampled. It is unclear whether this factor had an impact on the results, but negative removals were recorded 12 out of 18 times on these combined sites. In addition, the sites that did not have major parking issues had the lowest percentage of negative removals (zero of three or one of three sampling events).

The parking issues during sweeping events and/ or during post-sweeping sampling events for the six sites that had the highest percentage of negative removals (two of three sampling events) are listed below:

**Glendale**

- KIRK (SFR) – Sweepers had poor access to curb due to parking (not posted).
- KENN (SFR) – Sweepers had poor access to curb due to parking (not posted).
- ORNGE (COM) – Sweepers had poor access to curb due to parking (not posted).
- VERD (ART) – Sweepers had poor access to curb due to parking (not posted) and parking was an issue when sampling.

**Burbank**

- HOLL (ART) - Parking was an issue when sampling, posted no parking on sweeping day.
- MAG (COM) - Parking was an issue when sampling, posted no parking on sweeping day.

The average street dirt removals in the City of Burbank single family residential sites, CLARK and JEFF, were 13% and 25% respectively (**Table 7**). These positive removals reflect the fact that Burbank posts its streets as no parking for a two-hour period the day the sweeping event is scheduled to occur. The City of Glendale's industrial site (GRAND) was the only site in the City with all three of the sweeping events registering positive removals. The site is swept at night so parked cars do not seem to be an issue and that likely explains why the sweeping results showed the highest average pick-up of 43%.

### 3.5.2 Particle Size Distribution

As discussed in **Section 3.4.2**, the PSD data show that there were coarser materials in the pre-sweeping samples compared to the post-sweeping samples. The shift in materials means that on average, the street sweeping operations in the Cities are removing street dirt particles greater than 600 microns more effectively than those less than 600 microns.

### 3.5.3 Metals Concentrations

The highest metals concentrations were generally found in either the IND or COM land uses. In addition, the majority of the metals concentrations were found in the Fine and Medium particle size groups. For example, 94.5% of the cadmium found in the ART land use is associated with the Fine or Medium particle size groups (< 63 microns to 250 microns). This trend holds true for all metals in all land use types (**Table 11**). As the Cities' current street sweeping programs are more effective at removing materials greater than 600 microns, the Cities are not currently removing metals as effectively as possible.

**Table 11. Percent of Metals by Particle Size Group for all Land Uses in both Cities[1]**

| Metal | Particle Size Group | ART | COM | IND | SFR |
|---|---|---|---|---|---|
| Cadmium (Cd) | Fine & Medium (< 63 – 250 microns) | 94.5% | 93.0% | 90.9% | 88.8% |
| | Coarse (> 250 microns) | 5.5% | 7.0% | 9.1% | 11.2% |
| Copper (Cu) | Fine & Medium (< 63 – 250 microns) | 87.3% | 87.5% | 69.5% | 88.3% |
| | Coarse (> 250 microns) | 12.7% | 12.5% | 30.5% | 11.7% |
| Lead (Pb) | Fine & Medium (< 63 – 250 microns) | 87.2% | 87.4% | 84.8% | 88.5% |
| | Coarse (> 250 microns) | 12.8% | 12.6% | 15.2% | 11.5% |
| Zinc (Zn) | Fine & Medium (< 63 – 250 microns) | 90.7% | 90.8% | 72.9% | 85.9% |
| | Coarse (> 250 microns) | 9.3% | 9.2% | 27.1% | 14.1% |

1 – Percent of metals based on average concentrations.

### 3.5.4 Annual Sediment and Metals Mass Pick-Up by Site and by Land Use

The average annual mass pick-ups for sediment and metals by site (**Table 12**) were calculated using the mass of pre-sweeping sediment, the positive street dirt pick-up efficiencies from **Table 7**, the average metal concentrations from **Table 10**, and by using the existing sweeping frequencies. The equations used to calculate the annual sediment and metals mass pick-up by site are:

Annual Sediment Load Removed = Average Mass of Sediment Removed[1] x Annual Number of Sweeping Events

1 Does not include negative results

Annual Metals Load Removed = Annual Sediment Load of Fine Particles x Concentration on Fine Particles + Annual Sediment Load of Medium Particles x Concentration on Medium Particles + Annual Sediment Load of Coarse Particles x Concentration on Coarse Particles

For the City of Burbank, the most sediment is currently being removed from the CLARK site (SFR); the most cadmium and copper are removed from the WIN site (IND); and the most lead and zinc are removed from the MAG site (COM). For the City of Glendale, the most sediment is currently removed from the KENN site (SFR); the most cadmium is removed from the KENN site (SFR) and the ORNGE site (COM); and the most copper, lead, and zinc are removed from the GRAND site (IND). Similar to the results of the average metals concentrations by particle size (**Table 10**), the total mass of copper that is being removed from the IND land use in both Cities is significantly higher than the other land uses. Copper is the only metal where such a substantial distinction between land uses is evident.

**Table 12. Annual Mass Pick-Up of Street Dirt and Metals by Site. Bold values indicate the site with the most mass removed per metal/sediment.**

| City | Site ID | Land Use | Curb Mi | Sweep Program | Sweeps / Year | Mass Removed per Year from Each Site | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Sed (tns/yr) | Cd (lbs/yr) | Cu (lbs/yr) | Pb (lbs/yr) | Zn (lbs/yr) |
| Burbank | CLARK | SFR | 2.02 | Weekly (T/W) | 52 | **3.71** | 0.0022 | 0.288 | 0.185 | 0.981 |
| | JEFF | SFR | 1.38 | Weekly (W/Th) | 52 | 2.69 | 0.0011 | 0.154 | 0.097 | 0.504 |
| | MAG | COM | 1.0 | Weekly (T/F) | 52 | 2.92 | 0.0024 | 0.709 | **0.238** | **1.155** |
| | WIN | IND | 1.32 | Weekly (Th) | 52 | 1.30 | **0.0053** | **2.968** | 0.106 | 0.793 |
| | HOLL | ART | 1.0 | Weekly (W) | 52 | 0.93 | 0.0007 | 0.147 | 0.101 | 0.359 |
| Glendale | KENN | SFR | 1.6 | Bi-Monthly (M) | 26 | **3.31** | 0.0020 | 0.278 | 0.175 | 0.881 |
| | KIRK | SFR | 1.32 | Bi-Monthly (T) | 26 | 0.33 | 0.0000 | 0.011 | 0.007 | 0.041 |
| | ORNGE | COM | 1.34 | Weekly (M/W/F/Su) | 208 | 2.23 | 0.0020 | 0.525 | 0.189 | 1.001 |
| | GRAND | IND | 1.51 | Bi-Monthly (Tu) | 26 | 2.88 | **0.0106** | **5.532** | **0.209** | **1.488** |
| | VERD | ART | 1.0 | Bi-Monthly (W) | 26 | 0.49 | 0.0002 | 0.045 | 0.032 | 0.104 |

The average annual mass pick-up for sediment and metals by land use extrapolated for all of the monitored land uses for both Cities was calculated using the average sediment and metals masses removed by land use type (**Table 12**), the curb lengths of the monitoring sites (**Table 4**), and the estimated City-wide total curb miles for each monitored land use (**Table 13**).

**Table 13. City-wide Land Use Areas and Curb Miles for Land Uses Monitored**

| City | Land Use | Curb Miles |
|---|---|---|
| Burbank | SFR | 249.5 |
| | COM | 163.9 |
| | IND | 45.1 |
| | ART | 44.4 |
| Glendale | SFR | 335.3 |
| | COM | 129.5 |
| | IND | 46.6 |
| | ART | 35.7 |

For the City of Burbank, the most sediment, cadmium, copper, lead, and zinc are removed from the COM land uses throughout the City. For the City of Glendale, the most sediment, lead, and zinc are removed from the SFR land uses. However, the amount of zinc removed from the COM land use and the SFR land use are very similar: 96.7 and 97.6 pounds, respectively. The most cadmium and copper are removed from the IND land uses (**Table 14**). In the City of Burbank,

the mass of copper being removed from both the COM and IND land uses are notably higher compared to the other metal/land use combinations. In the City of Glendale, the mass of copper removed from the IND land use is the highest for all land uses in both Cities.

**Table 14. Annual Mass Pick-Up of Street Dirt and Metals by Land Use. Bold values indicate the land use with the most mass removed per metal/sediment.**

| | | | | Mass Removed per Year from Each Land Use | | | | |
|---|---|---|---|---|---|---|---|---|
| City | Land Use | Curb Mi | Sweeps / Year | Sed (tns/yr) | Cd (lbs/yr) | Cu (lbs/yr) | Pb (lbs/yr) | Zn (lbs/yr) |
| Burbank | SFR | 249.5 | 52 | 472 | 0.235 | 31.7 | 20.2 | 106.1 |
| | COM | 163.9 | 52 | **480** | **0.388** | **116.2** | **39.1** | **189.4** |
| | IND | 45.1 | 52 | 44 | 0.181 | 101.5 | 3.6 | 27.1 |
| | ART | 44.4 | 52 | 42 | 0.026 | 6.5 | 4.5 | 16.0 |
| | | | Total per Year: | 1,038 | 0.83 | 255.9 | 67.4 | 338.6 |
| Glendale | SFR | 335.3 | 26 | **389** | 0.220 | 30.5 | **19.3** | **97.6** |
| | COM | 129.5 | 208 | 215 | 0.183 | 50.8 | 18.3 | 96.7 |
| | IND | 46.6 | 26 | 89 | **0.328** | **170.7** | 6.5 | 45.9 |
| | ART | 35.7 | 26 | 18 | 0.007 | 1.6 | 1.1 | 3.7 |
| | | | Total per Year: | 711 | 0.74 | 253.6 | 45.2 | 243.9 |

# 4 Phase II: Potential Enhancements - Controlled Environment

Phase II focused on evaluating the effect of potential enhancements, including sweeper type and sweeper speed, on street dirt pickup effectiveness in a controlled environment. Phase II testing was conducted with sweepers currently utilized by the Cities at two operating speeds to evaluate the influence of sweeper type and speed on efficiency. Phase II data collection efforts aid in evaluating both the effects of sweeper type and driving speeds which supports the analysis of how these factors, along with sweeping frequency and potential interference from parked cars, influence the effectiveness of the current programs.

## 4.1 APPROACH

Five sweeper types from the Cities current fleet of street sweepers were tested under two operating speeds (5 mph and 10 mph) to evaluate the effects of sweeper type and speed on pickup effectiveness. The sweeper types tested were mechanical, vacuum, and regenerative air (vacuum and regenerative air are both considered to be air machines). In addition, two initial street dirt simulant loadings were used during Phase II testing, three pounds and six pounds, to determine pickup effectiveness under two loading conditions representative of actual conditions. The simulant was a mixture of various silica based products combined to closely match the specific gravities and PSD to those of the street dirt collected during Phase I pre-sweeping sampling. Matching the simulant PSD to the PSD of the street dirt collected during Phase I sampling ensures that the potential performance enhancement tests were conducted using

variables that represent conditions found in the field. **Table 15** lists the sweeper types and the tests performed during Phase II.

**Table 15. Sweepers Tested and the Tests performed during Phase II**

| City | Sweeper Type | 5 mph/ 3 lbs | 10 mph/ 3 lbs | 5 mph/ 6 lbs | 10 mph/ 6 lbs |
|---|---|---|---|---|---|
| Burbank | Schwarze M6000[1] | X | X | X | X |
| | Schwarze A7000[2] | X | X | X | X |
| Glendale[3] | Elgin Crosswind[2] | X | X | X | X |
| | Elgin Broome Bear[1] | | X | X | X |
| | Allianz CNG 4000[1] | X | | X | X |

1 – Mechanical sweeper.
2 – Air sweeper.
3 – Since the City of Glendale tested three sweeper types, the Elgin Broom Bear and Allianz CNG 4000 were not tested using the variables 5 mph/ 3 lbs and 10 mph/ 3 lbs respectively.

Phase II testing had to be conducted in a controlled environment because the initial loading and PSD must be known so when the remaining material is hand vacuumed and sieved, the sweeper's pick-up performance by particle size can be accurately determined. The location for the Phase II tests was the City of Burbank Water Reclamation Plant. This location was chosen because it is a controlled environment that has pavement similar to the pavement seen in the sampling areas without any large cracks, the area is protected from the wind, and there was no need for traffic control. Before the test material (simulant) was applied, the street sweeper operator was given the opportunity to practice several timed sweeping passes to ensure that the operator could successfully execute the desired sweeping speed when the actual test was performed. These practice runs also ensured that the track was very clean before the simulant was applied. Additionally, the track was hand-vacuumed prior to the first test.

## 4.2 SAMPLE COLLECTION

For each test, a known quantity of the street dirt simulant was spread evenly along a 50 foot (ft) section of a curb-lined track at a width of approximately two feet (**Figure 7**). The street sweepers performed a single pass at a specified forward speed with a known loading (**Figure 8**). The actual time that the machine spent cleaning the test track was recorded using a stopwatch and the average sweeping speed was calculated.

**Figure 7. Applying the Simulant**

**Figure 8. Sweeper Performing Phase II Test**

Following sweeping, the test track was allowed to dry, since water was used for fugitive-dust control in all but one of the sweeper tests. Due to the warm temperatures and sunny conditions, the track dried in approximately five to ten minutes. Once the track was dry, all of the remaining simulant on the test track, and any simulant displaced from the original 50 ft by 2 ft application area by the gutter broom, was collected using a stainless steel Shop-Vac® fitted with a Dacron filter bag to keep any fine materials from entering the filter/motor unit (**Figure 9**). The sample transfer and storage procedures were the same as those performed in Phase I. In addition, during Phase II, the samples were weighed on site and compared to the original loading amount to get an estimate of pickup effectiveness.

**Figure 9. Phase II Sample Collection**

## 4.3 ANALYTICAL METHODS

The samples collected during Phase II, as well as a sample of the simulant, were analyzed for PSD. The analytical methods for PSD were the same as in Phase I. Each sample was weighed in the laboratory and then dried in an oven kept below 120° F. Each sample was then sifted through a series of seven stainless steel sieves with various sizes. The fractions that passed through each sieve were weighed separately to determine the PSD. **Table 6** lists the sieve sizes and the corresponding particle size range in microns used in the PSD analyses.

## 4.4 RESULTS

Overall pick-up efficiency for each sweeper test was calculated as a percentage of the initial material removed. The calculation was based on the weight of remaining material collected following the sweeping operation compared to the known weight of simulant spread evenly along the test track. The pick-up performance results are presented in **Table 16**. Pick-up efficiency in individual tests ranged from 28% (Schwarze M6000) to 84% (Elgin Crosswind). The average efficiency ranged from 55.5% (Schwarze M6000) to 79.4% (Elgin Crosswind). On average, the two most efficient sweepers were the Schwarze A7000 and the Elgin Crosswind. In addition, the average pick-up efficiency was generally higher for the 10 mph tests compared to the 5 mph tests. However, there did not seem to be a noticeable difference in pick-up effectiveness between the two loading rates when the sweepers were operated at the same speed.

A sieve analysis was also conducted on each of the 18 Phase II samples as well as on a single sample of the simulant to determine the PSD to estimate the sweeper's pick-up efficiencies for each of the eight particle size groups used in the sieve analysis. The data showed that the sweepers were more effective at removing materials greater than 250 microns than they were at removing materials less than 250 microns. PSD results from the sieve analysis are presented in **Appendix C**.

**Table 16. Phase II Street Sweeper Test Results. Bold values indicate the sweepers with the highest percent removal.**

| City | Sweeper | Targeted Speed (mph) | Actual Speed (mph) | Initial Loading (lbs/ curb mi) | Post-Sweeping Loading (lbs/ curb mi) | Removal (%) | Average % Removal | Test Notes |
|---|---|---|---|---|---|---|---|---|
| Burbank | Schwarze M6000 | 5 | 4.55 | 634 | 175 | 72 | 55.5 | Flat broom, no pitch, with water |
| | Schwarze M6000 | 5 | 5.41 | 317 | 229 | 28 | | Flat broom, no pitch, with water |
| | Schwarze M6000 | 10 | 9.58 | 634 | 225 | 65 | | Flat broom, no pitch, with water |
| | Schwarze M6000 | 10 | 10.49 | 317 | 136 | 57 | | Flat broom, no pitch, with water |
| | Schwarze A7000 | 5 | 5.3 | 634 | 188 | 70 | 77.7 | With water |
| | Schwarze A7000 | 5 | 4.08 | 317 | 56 | **82** | | Slight breeze, flat broom, no water |
| | Schwarze A7000 | 10 | 10.79 | 634 | 118 | **81** | | Slight breeze, flat broom, no water |
| Glendale | Elgin Crosswind | 5 | 7.07 | 634 | 155 | 76 | 79.4 | With water, gutter broom pitched 5% |
| | Elgin Crosswind | 5 | 7.12 | 634 | 101 | **84** | | Flat broom, no pitch, with water |
| | Elgin Crosswind | 5 | 6.74 | 317 | 69 | 78 | | Flat broom, no pitch, with water |
| | Elgin Crosswind | 10 | 10.39 | 634 | 132 | 79 | | Flat broom, no pitch, with water |
| | Elgin Crosswind | 10 | 11.18 | 317 | 62 | 80 | | Flat broom, no pitch, with water |
| | Allianz CNG 4000 | 5 | 5.4 | 634 | 194 | 69 | 72.7 | Flat broom, no pitch, with water |
| | Allianz CNG 4000 | 5 | 5.29 | 317 | 95 | 70 | | Flat broom, no pitch, with water |
| | Allianz CNG 4000 | 10 | 10.79 | 634 | 133 | 79 | | Flat broom, no pitch, with water |
| | Elgin Broome Bear | 5 | 4.96 | 634 | 160 | 75 | 68.7 | Flat broom, no pitch, with water |
| | Elgin Broome Bear | 10 | 9.66 | 634 | 184 | 71 | | Slight breeze, no pitch, with water |
| | Elgin Broome Bear | 10 | 10 | 317 | 126 | 60 | | Flat broom, no pitch, with water |

## 4.5 DISCUSSION

The results from the Phase II tests were used to ascertain which sweepers were the most effective via potential enhancements in a controlled environment as discussed below.

### 4.5.1 Pick-Up Effectiveness – Fugitive Dust Loss

During Phase II, the sweeper tests were videotaped to provide visual evidence of each test. Qualitative analysis of the videos taken during each sweeper test indicated that fugitive dust losses were visible during each of the tests. The relative amount of fugitive dust loss was categorized qualitatively as low, medium or high. The losses to fugitive dust from the Elgin Crosswind regenerative air appeared to be the lowest compared to all the other sweepers. The Alliance 4000 mechanical appeared to have the highest amount of fugitive dust loss. This puts the 73% average overall measured pick-up for the Alliance into question, especially during the 10 mph test where the video shows a significant cloud of dust and the test recorded a 79% pick-up.

Another machine that exhibited a significant amount of fugitive dust loss was the Schwarze A7000 vacuum sweeper. It is difficult to estimate how much of the 77.7% average pick-up was actually lost to fugitive dust. Wind was a factor during the third test of the A7000 which measured an 82% pick-up efficiency. The other two mechanical sweepers, Schwarze M6000 and the Elgin Broom Bear, both exhibited a moderate level of fugitive dust loss. This might be explained by the higher amount of water spray the operators of these two machines applied. The amount of water used during the test would be difficult to sustain during regular sweeping.

### 4.5.2 Pick-Up Effectiveness – Performance of Existing City Sweepers

The pick-up performance results, based on percent removal, for the Schwarze M6000 and the Elgin Broom Bear were considered fair to good. The Schwarze M6000, used by Burbank, averaged a 56% overall pick-up for the four tests whereas the Elgin Broom Bear, used by Glendale, averaged 69% pick-up for only three tests. The wind, which was not a factor for the first two days of testing, started to increase in the early afternoon of day three (second and third test of the Broom Bear) and it increased during the late morning of day four (third test of the Schwarze A7000). The wind influence could explain some of the surprising results for the Broom Bear. The one test that was not run on the Broom Bear (i.e. loading of 317 lbs per curb mile at 5 mph) was previously run for the M6000 and it recorded only a 28% pick-up which was the lowest pick-up performance recorded for any of the 18 tests.

The other M6000 tests showed that, at a given speed such as 10 mph, the pick-up performance decreases with decreasing loading. Note that the pick-up was reduced from 65% for 634 lbs per curb mile down to 57% for 317 lbs per curb mile. This is an expected result. Note that when the M6000 slowed down while sweeping the higher loading of 634 lbs per curb mile the pick-up performance increased to 72% from its previous 65%. This also was an expected result since pick-up performance usually increases for slower speeds while sweeping the same loading. Therefore, for the lower loading of 317 lbs per curb mile swept at the lower speed test, some decrease from the previous 65% would be expected when the loading is reduced. But the decrease should have been offset by the fact that the machine slowed down. Based on the other tests performed with this sweeper during Phase II of the Pilot Program, the expected performance was around 60%. Instead the recorded removal dropped to 28%. This was lower

than the 57% the M6000 picked up at 10 mph for the same 317 lbs per curb mile loading. It is therefore quite likely that there was a problem with the operation of the machine during this test. Perhaps the gutter broom was not lowered enough during that test since the video did show that a lot of material seemed to remain on the pavement. If this result is excluded, the average pick-up for the M6000 is 65%, which is comparable to the 69% average for the Broom Bear mentioned previously.

The average pick-up performance, based on percent removal, of the Elgin Crosswind was the best with an overall pick-up of 79.4%. However, if the first test result of 76% is excluded from the average pick-up calculation, then the overall average for the Crosswind improves slightly to 80%. During the first test, the machine's gutter broom was tilted at 5% instead of laying flat. When the same test was repeated with the gutter broom flat (both tests had a loading of 634 lbs per curb mile and a targeted speed of 5 mph), the pick-up improved to 84%. When the loading was decreased to 317 lbs per curb mile at the same targeted speed of 5 mph, the pick-up decreased to 78%. The loading was then increased to the 634 lbs per curb mile and the speed was increased to 10 mph and water was not used for fugitive dust suppression and the pick-up essentially remained the same at 79%.

# 5 Phase III: Field Test of Potential Enhancements

Phase III focused on determining the pickup effectiveness in the field based upon implementing the potential performance enhancements evaluated under Phase II at a subset of sites evaluated in Phase I.

## 5.1 APPROACH

During Phase III, street dirt samples were collected at one residential area in each city (Burbank: Jeffries Ave; Glendale: W Kenneth Rd) and the industrial area in Burbank (Winona Ave). Phase III testing was similar in approach to Phase I testing; however, the sweeper operators were aware of the testing. The sweepers chosen for Phase III sampling were the Schwarze A7000 and Elgin Crosswind as these two sweepers had the best pick-up efficiency during Phase II. In addition, the sweepers were operated at 10 mph as Phase II results showed comparable pick-up efficiency at 10 mph and 5 mph, with 10 mph still providing reasonable sweeping coverage during an average work day.

Phase III sampling was conducted on days when the areas were not scheduled to be swept in order to ensure the street dirt was representative of typical conditions. As in Phase I, five 10 ft by 10 ft areas (500 sq ft total) in each of the three sites were sampled prior to street sweeping and five different 10 ft by 10 ft areas at each of the three sites were sampled after street sweeping. However, in contrast to Phase I, the pre-sweep sampling and the post-sweep sampling occurred on the same day as the sweeper operators were participants in this Phase of testing.

The areas that were sampled (5 pre-sweep and 5 post-sweep) within each of the three sampling sites were chosen at random, but at locations where the sweepers would have access to the curb. Example locations include areas where parking is not allowed (i.e., red-painted curbs) or in locations where parked cars were not a concern and there was ample curb area for the sweepers to access.

## 5.2 SAMPLE COLLECTION

Sample collection methods were the same as in Phase I (**Section 3.2**).

## 5.3 ANALYTICAL METHODS

Samples collected during Phase III were analyzed for PSD using the same methodologies as those used in Phase I and Phase II (**Section 3.3 and Section 4.3**). Samples were not analyzed for metal concentrations.

## 5.4 RESULTS

The overall pick-up efficiency of the sweepers tested during Phase III ranged from 15% to 61%. The pick-up efficiency results from the Phase III street sweeping field tests are presented in **Table 17**. In addition, the PSD results from Phase III are provided in **Appendix D**.

**Table 17. Phase III Street Sweeping Field Test Results**

| City | Site ID | Land Use | Sweeper Type | Total Loading Pre Sweep (lbs/ curb mile) | Total Loading Post Sweep (lbs/ curb mile) | % Removal |
|---|---|---|---|---|---|---|
| Burbank | JEFF | SFR | Schwarze A7000 | 130 | 110 | 15 |
| | WIN | IND | Schwarze A7000 | 169 | 93 | 45 |
| Glendale | KENN | SFR | Elgin Crosswind | 250 | 96 | 61 |

## 5.5 DISCUSSION

The effectiveness of the potential modifications (high efficiency sweepers, forward speed of 10 mph, access to the curb) are discussed both at the site level (**Section 5.5.1**) and extrapolated to the citywide scale for both Cities (**Section 5.5.2**).

### 5.5.1 Tested Sites

Street dirt pick-up performance increased from -10% (Phase I) to 61% (Phase III) at W Kenneth Rd (SFR) in Glendale. This site was swept by the Elgin Crosswind that had access to the curb. The pre-sweeping loading was only 250 lbs per curb mile, which is the lowest pre-sweeping loading measured at the site throughout the Pilot Program (pre-sweeping loading from Phase I ranged from 307 to 608 lbs per curb mile). Even with such a low pre-sweeping load, the sweeper still managed to remove 153 lbs per curb mile and reduce the post-sweeping load down to 96 lbs per curb mile.

Street dirt pick-up performance at Winona Ave (IND) in Burbank also greatly improved from 12% (Phase I average) to 45% (Phase III). This site was swept by the Schwarze A7000 sweeper. The four fold increase in pick-up efficiency was a direct result of switching to the Schwarze A7000 vacuum verses the M6000 or the Alliance 4000 mechanical sweeper used in the Phase I tests as interference from parked cars was not a problem at this site during Phase I. Similar to the W. Kenneth Rd site in Glendale, the pre-sweeping load of 169 lbs per curb mile was the lowest measured throughout the Pilot Program (pre-sweep load from Phase I ranged from 227 to 273 lbs per curb mile). The vacuum sweeper still managed to pick-up 76 lbs per curb mile of street dirt and reduce the post-sweeping load down to 93 lbs per curb mile. The results for W

Kenneth Rd and Winona Ave indicate that if a high performance machine, such as the Elgin Crosswind or Schwarze A7000, has access to the curb, it has the potential to substantially remove more street dirt compared to a mechanical sweeper, even while traveling at 10 mph. However, the Phase II tests showed that fugitive dust loss may be an issue for the Schwarze A7000. Such losses can be reduced, potentially by decreasing the forward speed. Phase II results indicated that the Schwarze A7000 can be operated at a slower speed (5 mph) and achieve similar pick-up performance.

The results from Jeffries Ave (SFR) in Burbank did not show an increase in pick-up performance, decreasing from 25% (Phase I) to 15% (Phase III). However, the pre-sweeping load was only 130 lbs per curb mile, which is 123 lbs per curb mile less than the lowest Phase I pre-sweeping load measured at the site (JEFF pre-sweeping loading from Phase I ranged from 253 to 290 lbs per curb mile). Even with such a low pre-sweeping load, the Schwarze A7000 managed to pick-up 20 lbs per curb mile, reducing the loading to 110 lbs per curb mile. The results at Jeffries Ave (SFR) illustrate that there is a minimum threshold of street dirt accumulation below which the sweeper will not remove much or any of the accumulated loading. This minimum threshold is a function of the type of sweeper being used and varies by the particle size group.

The key findings from Phase III demonstrate that if the street sweeper can get to the curb where access was previously limited, the pick-up performance can increase substantially. Additionally, if regenerative air or vacuum machines are used instead of mechanical sweepers, then the pick-up performance increases due to the air machines inherent ability to pick-up a greater amount of the finer street dirt.

### 5.5.2 Potential Effectiveness of Enhancements - Citywide

This section extrapolates the results of Phase III to the entire Cities, by land use. The intent is to provide estimates of potential increased removal of metals through enhancement of the existing sweeping program. The estimates are intended to inform management decisions related to attaining the WLAs for the Metals TMDL.

The methodologies used to calculate metal removal estimates are similar to the methodology employed during the Phase I data analyses (Section 3.5). Three different approaches were taken to estimate the mass of sediment and metals removed. The first assumed that the highest percent removal achieved during the controlled testing conducted in Phase II using an initial loading most similar to initial loadings observed during Phase I could be achieved in all land uses. The second assumed that the highest percent removals achieved during the Phase III could be achieved in all land uses. Note that percent removal can lead to misleading calculations when initial loadings are low and pickup performance may be limited or alternatively initial loadings are high and performance can be overstated. However, the Phase II highest percent removal used an initial loading similar to the loading rates observed in the field during Phase I. Additionally, the Phase III initial loadings with the highest percent removal were somewhat lower than those observed in the field. This reduces the potential for overestimating percent removal. The third approach assumed that the post-sweep load was reduced to the post-sweep loads achieved during Phase III. This approach essentially assumes that below a certain level of street dirt load the sweeper can no longer remove sediment.

In the process of doing the analysis, the extremely low pre-sweep load measured at Orange St (COM) in Glendale, resulting from the four times per week sweeping frequency, appeared to

artificially lower the metals pick-up estimate for the COM land use in an enhanced sweeping program in Glendale. As discussed in Section 3.5.1, low street dirt accumulations associated with more frequent sweeping appear to make detecting differences between pre and post-sweep loads difficult. As such, in order to more accurately reflect the COM land use in Glendale, the pre-sweep load from Magnolia Ave (COM) in Burbank was used instead of the pre-sweep load measured at Orange St (COM) in Glendale. In addition, the frequency of sweeping for the COM land use in Glendale was adjusted to weekly, consistent with the sweeping frequency of the Magnolia Ave (COM) site in Burbank. As there is a relationship between sweeping frequency and pre-sweep loads, the frequency associated with the pre-sweep load data from Magnolia Ave (COM) must be maintained in order for the metals pick-up estimation to remain appropriate.

For the two approaches that utilized the highest percent removals achieved during Phase II and Phase III testing, the potential metals pick-up by an enhanced street sweeping program were based upon the following parameters:

- Percent Removal: For removal estimates based on the Phase II removal percentage, the highest percent removals achieved (Burbank Schwarze A7000 at 5 mph at an initial loading of 317 lbs per curb mile) was used for both Cities. For removal estimates based on the Phase III removal percentage the highest percent removals achieved (Elgin Crosswind at 10 mph) was used for both Cities.
- Pre-Sweep Load: Average load from each site measured under Phase I, with the exception of Orange Street (COM) in Glendale (see above).
- Post-Sweep Load: Calculated by multiplying the percent removal (either highest from Phase II or Phase III) for the various grain sizes by the pre-sweep load by grain sizes and summing.
- PSD: PSD of street dirt removed (calculated as part of determining the post-sweep load).
- Metals Concentrations: Metals concentrations measured on the three fraction groups in Phase I.
- Load of Metals Removed : Calculated by multiplying the post-sweep sediment load by grain size by the corresponding metals concentrations. These calculations were conducted in same manner as calculating loads removed under the current sweeping program (see Section 3.5.4).

The potential metals pick-up by an enhanced street sweeping program assuming the post-sweep load achieved during Phase III was based upon the following parameters:

- Street Sweeper Type: Elgin Crosswind is used to sweep all land uses in Glendale; Schwarze A7000 is used to sweep all land uses in Burbank (most effective sweepers based upon Phase II).
- Pre-Sweep Load: Average load from each site measured under Phase I, with the exception of Orange Street (COM) in Glendale (see above).
- Post-Sweep Load: Lowest post-sweep load for each City measured under Phase III.
- PSD: PSD of street dirt removed calculated by subtracting the post-sweep load from the pre-sweep load by grain size fraction.

- Metals Concentrations: Metals concentrations measured on the three fractions during tested in Phase I.

For the City of Burbank, the estimation assumed that every time a sweeping event occurred in Burbank, the post-sweep load was reduced to 93 lbs per curb mile [the post-sweep load achieved by the Schwarze A7000 vacuum sweeper in the Phase III field test at Winona Ave (IND)]. The removal amounts were calculated as the difference between the Phase I pre-sweep load minus the same post sweeping load for each land use (93 lbs per curb mile).

For the City of Glendale, the estimation assumed that every time a sweeping occurred in Glendale, the post-sweep load was reduced to 96 lbs per curb mile (the post-sweep load achieved by the Elgin Crosswind regenerative air sweeper in the Phase III field test at Kenneth Rd (SFR)). Similarly, the amount of material pick-up by the Glendale sweeping operation was calculated as the difference between the Phase I pre-sweep load minus the same post sweeping load for each land use (96 lbs per curb mile).

The potential effectiveness (annual removal) of implementing an enhanced program based upon the assumption of the highest percent removal measured during Phase II, the highest percent removal measured during Phase III, and the lowest post-sweep loading for each monitored land use in each City are presented in **Table 18**, **Table 19**, and **Table 20**, respectively. **Table 21** presents a comparison of the metal masses estimated for the current sweeping programs in each city and indicates that the enhanced program could potentially increase the removal of metals by anywhere from two to almost four fold. The estimates presented in **Table 18**, **Table 19**, **Table 20** and **Table 21** should be viewed as a potential range for the maximum effectiveness for a modified sweeping program, based upon the data and results of the Pilot Program.

**Table 18. Potential Effectiveness (Annual Removal) of an Enhanced Sweeping Program Based on the Highest Percent Removal in Phase II. Bold values indicate the land use with the highest potential mass removal for each metal/sediment.**

| City | Land Use | Total Curb Mi | Sweeps Per Yr | Mass Removed Per Year from Each Land Use | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | Sed (tons/yr) | Cd (lbs/yr) | Cu (lbs/yr) | Pb (lbs/yr) | Zn (lbs/yr) |
| Burbank | SFR | 249.5 | 52 | **1498** | 0.88 | 128 | 80 | 397 |
| | COM | 163.9 | 52 | 1412 | **1.40** | 453 | **147** | **659** |
| | IND | 45.1 | 52 | 223 | 0.82 | **500** | 18 | 129 |
| | ART | 44.4 | 52 | 161 | 0.11 | 26 | 18 | 65 |
| **Burbank: Total Per Year:** | | | | **3294** | **3.21** | **1107** | **263** | **1250** |
| Glendale | SFR | 335.3 | 26 | 1058 | 0.49 | 72 | 45 | 231 |
| | COM[1] | 129.5 | 52 | **1116** | **1.11** | **358** | **116** | **521** |
| | IND | 46.6 | 26 | 152 | 0.51 | 297 | 11 | 78 |
| | ART | 35.7 | 26 | 133 | 0.07 | 18 | 13 | 43 |
| **Glendale: Total Per Year:** | | | | **2459** | **2.17** | **745** | **184** | **872** |

1 = Based on pre-sweep load from W. Magnolia Ave (COM) in Burbank

**Table 19. Potential Effectiveness (Annual Removal) of an Enhanced Sweeping Program Based on the Highest Percent Removal in Phase III. Bold values indicate the land use with the highest potential mass removal for each metal/sediment.**

| City | Land Use | Total Curb Mi | Sweeps Per Yr | Mass Removed Per Year from Each Land Use | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | Sed (tons/yr) | Cd (lbs/yr) | Cu (lbs/yr) | Pb (lbs/yr) | Zn (lbs/yr) |
| Burbank | SFR | 249.5 | 52 | **1132** | 0.65 | 93 | 58 | 292 |
| | COM | 163.9 | 52 | 1066 | **1.00** | 323 | **106** | **477** |
| | IND | 45.1 | 52 | 165 | 0.60 | **370** | 13 | 96 |
| | ART | 44.4 | 52 | 121 | 0.08 | 19 | 13 | 48 |
| **Burbank: Total Per Year:** | | | | **2484** | **2.32** | **805** | **190** | **913** |
| Glendale | SFR | 335.3 | 26 | 810 | 0.36 | 53 | 33 | 171 |
| | COM[1] | 129.5 | 52 | **842** | **0.79** | **255** | **84** | **377** |
| | IND | 46.6 | 26 | 116 | 0.37 | 220 | 8 | 58 |
| | ART | 35.7 | 26 | 103 | 0.05 | 14 | 9 | 32 |
| **Glendale: Total Per Year:** | | | | **1871** | **1.57** | **542** | **134** | **638** |

1 = Based on pre-sweep load from W. Magnolia Ave (COM) in Burbank

**Table 20. Potential Effectiveness (Annual Removal) of an Enhanced Sweeping Program Based on Lowest Post-Sweep Load Observed in Phase III. Bolded values indicate the land use with the highest potential mass removal for each metal/sediment.**

| City | Land Use | Total Curb Mi | Sweeps Per Yr | Mass Removed Per Year from Each Land Use | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | Sed (tons/yr) | Cd (lbs/yr) | Cu (lbs/yr) | Pb (lbs/yr) | Zn (lbs/yr) |
| Burbank | SFR | 249.5 | 52 | 1267 | 0.807 | 113.6 | 71.4 | 355.4 |
| | COM | 163.9 | 52 | **1327** | **1.389** | **435.5** | **142.6** | **655.5** |
| | IND | 45.1 | 52 | 170 | 0.736 | 392.0 | 14.9 | 103.4 |
| | ART | 44.4 | 52 | 89 | 0.055 | 13.4 | 9.5 | 32.7 |
| | **Burbank: Total Per Year:** | | | **2853** | **2.987** | **954.5** | **238.4** | **1147.0** |
| Glendale | SFR | 335.3 | 26 | 871 | 0.362 | 52.1 | 32.7 | 172.2 |
| | COM[1] | 129.5 | 52 | **1036** | **1.043** | **327.3** | **107.7** | **497.0** |
| | IND | 46.6 | 26 | 137 | 0.507 | 252.7 | 9.8 | 68.7 |
| | ART | 35.7 | 26 | 114 | 0.046 | 13.6 | 9.3 | 30.7 |
| | **Glendale: Total Per Year:** | | | **2158** | **1.958** | **645.7** | **159.5** | **768.6** |

1 = Based on pre-sweep load from W. Magnolia Ave (COM) in Burbank

**Table 21. Comparison of Estimates on the Potential Effectiveness of an Enhanced Sweeping Program.**

| City | Loading Estimate | Mass Removed Per Year | | | | |
|---|---|---|---|---|---|---|
| | | Sed (tons/yr) | Cd (lbs/yr) | Cu (lbs/yr) | Pb (lbs/yr) | Zn (lbs/yr) |
| Burbank | Current | 1038 | 0.830 | 256 | 67 | 339 |
| | Highest Phase II % Removal | 3294 | 3.21 | 1107 | 263 | 1250 |
| | Highest Phase III % Removal | 2484 | 2.32 | 805 | 190 | 913 |
| | Lowest Phase III Post-Sweep Load | 2853 | 2.99 | 955 | 238 | 1147 |
| Glendale | Current | 711 | 0.738 | 254 | 45 | 244 |
| | Highest Phase II % Removal | 2459 | 2.17 | 745 | 184 | 872 |
| | Highest Phase III % Removal | 1871 | 1.57 | 542 | 134 | 638 |
| | Lowest Phase III Post-Sweep Load | 2158 | 1.96 | 646 | 160 | 769 |

# 6 Recommendations for Enhanced Street Sweeping Program

Based upon the results of the Pilot Program, recommendations to be considered by the Cities are provided below. These recommendations, in whole or in part, are intended to enhance the existing sweeping programs and increase the removal of metals associated with street dirt. As noted in **Section 7**, City staff will need to evaluate these recommendations, combined with all other necessary factors, and select, as appropriate, the recommendations that will be implemented as the basis for an Enhanced Street Sweeping Program.

1. **Consider posting streets in the City of Glendale**. The City of Glendale should consider conducting an assessment of the cost of posting appropriate streets as "no parking" and enforcement of parking restrictions during sweeping operations. Posting of streets and enforcement of parking restrictions would enhance the ability for sweepers to access the curb.
2. **Consider assessing effectiveness of parking restrictions**. Enforcement of parking restrictions is directly related to the effectiveness of posting streets. The City of Burbank currently enforces parking restrictions on streets posted "no parking" during sweeping operations while the City of Glendale may need to assess enforcement options if new streets are posted (see Recommendation 1). An assessment of methods to increase enforcement may potentially aid in increasing access to the curb.
3. **Consider adjusting sweeper fleets**. The Cities currently operate mechanical and regenerative air machines. As tested in Phase II, certain machines are achieving higher removal rates than others (Schwarze A7000 and Elgin Crosswind); however, an enhanced sweeping program could focus on reducing fugitive dust loss and increasing the removal of particles less than 250 microns. One method of attaining both goals would be to consider adjusting the sweeper fleets to incorporate more high efficiency machines available from the three major manufacturers, Elgin, Schwarze, and Tymco. To date, a study has not been conducted to evaluate the performance of high efficiency models as they have only become commercially available over the last several years. However, a previous study in 1997 evaluated a sweeper model (Sutherland and Jelen, 1997) and the sweeping technology of that model surpassed even that of the regenerative air sweepers that were available at the time. This model is no longer available but the original company, Enviro Whirl Technologies, was acquired by Schwarze Industries.

   Alternatively, the Cities could utilize the results of Phase II to replace the lowest performing machines with the vacuum/regenerative air machines that achieved the best removal rates. In addition to this Pilot Program, the effectiveness of Elgin machines were evaluated in 2009 (Sutherland 2009).

Alternatively, the Cities could utilize the results of Phase II to replace the lowest performing machines with the vacuum/regenerative air machines that achieved the best removal rates. In addition to this Pilot Program, the effectiveness of Elgin machines were evaluated in 2009 (Sutherland 2009).

4. **Consider developing and implementing targeted forward sweeping speeds**. In evaluating potential improvements to the existing programs, forward sweeping speed should be evaluated. Everything else considered equal, the pickup effectiveness increases as the forward speed decreases and there is a loss of efficiency above 10 mph (Sartor and Boyd 1972). However, in this Pilot Program, both 5 mph and 10 mph were evaluated as potential forward operating speeds and both speeds performed relatively equally well. Based upon the results of Phase II, a forward speed between 5-10 mph should be considered for an enhanced sweeping program.
5. **Consider prioritizing certain land uses as enhancements are phased into the programs**. As certain improvements may occur over a period of several years (e.g., the purchase of new machines), consider targeting certain land uses as the enhancements are phased in to achieve the highest immediate benefit. For example, targeting COM and/or IND land uses in the City of Burbank could substantially increase metals removal (in particular, copper) by the sweeping program.
6. **Consider reviewing sweeping frequency**. The frequency of sweeping should be considered in order to maximize the benefit (metals removed) while minimizing the costs. Specifically, the results for Orange Street from Phase I seem to indicate that sweeping four times per week may not be an effective use of sweeping resources in terms of street dirt removal. Increased removal rates could be achieved with significantly less sweeping events.
7. **Consider using technology to support effective sweeping practices**. GPS technology can support effective sweeping practices by assessing actual versus targeted forward sweeping speed.
8. **Consider providing opportunities for operators to provide feedback**. Operators have on the ground knowledge of the challenges associated with conducting effective sweeping. Operator input would provide insight on operational challenges and modifications to the sweeping program.
9. **Consider reviewing maintenance and standard operating procedures (SOPs)**. In order to maintain the street dirt removal efficiencies of the Cities' sweeper fleets, evaluate current maintenance practices and SOPs to ensure they are consistent with the manufacturer's recommendations. In particular, as evident in the Phase II testing, the pitch of the gutter broom can influence effectiveness of street dirt removal. Assessment of SOPs could include ensuring the pitch of the gutter broom is appropriately set for the given conditions.
10. **Consider evaluating potential interference from garbage collection days in residential areas**. Similar to parked car interference, a street lined with garbage cans would restrict access to the curb. In making any adjustments to the sweeping schedule, consideration should be given to garbage collection days to continue to avoid overlapping schedules.

# 7 Next Steps

Consistent with the Cities' Implementation Plan, the Cities will need to evaluate the recommendations presented in Section 6 and implement enhancements to the street sweeping

programs as appropriate. An evaluation of the effectiveness of the enhanced program is scheduled for September 2015.

The next steps to implement these enhancements include:

- **Develop Enhanced Sweeping Program**. City staff in various appropriate departments should review the recommendations in this Pilot Program report. Considering all necessary factors (TMDL obligations, logistical constraints, financial considerations, etc.) an enhanced sweeping program should be developed. The scale of implementation (e.g., citywide, targeted land uses, targeted metals source areas) should also be identified.
- **Implement Enhanced Sweeping Program**. The enhanced program should be implemented at the selected scale for each City.
- **Evaluate Effectiveness of the Enhanced Sweeping Program in Meeting Phase I goals.** Consistent with the Implementation Plan, an evaluation of the effectiveness of the enhanced sweeping program is due to the Regional Board in September 2015. The Cities should develop the method for evaluating the enhanced sweeping program, conduct the assessment, and submit the results to the Regional Board.

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# Appendix A – City of Burbank Phase I Pre-Sweeping and Post-Sweeping Data

| Site ID | Land Use | Event | Sample ID | Date | Time | Sweeping Type | Total Dry Accumulation | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Mass (g) | Loading (g/curb m) | Loading (lbs/curb mi) |
| HOLL | ART | 1 | 1-holl-pre | 3/8/2011 | 14.45 | PRE | 971 | 63.7 | 226 |
| HOLL | ART | 1 | 1-holl-post | 3/10/2011 | 11.20 | POST | 564 | 37.0 | 131 |
| HOLL | ART | 2 | 2-holl-pre | 4/12/2011 | 10.30 | PRE | 685 | 44.9 | 159 |
| HOLL | ART | 2 | 2-holl-post | 4/14/2011 | 10.30 | POST | 692 | 45.4 | 161 |
| HOLL | ART | 3 | 3-holl-pre | 5/10/2011 | 10.30 | PRE | 536 | 35.2 | 125 |
| HOLL | ART | 3 | 3-holl-post | 5/12/2011 | 10.30 | POST | 734 | 48.2 | 171 |
| MAG | COM | 1 | 1-mag-pre | 4/4/2011 | 11.00 | PRE | 1843 | 120.9 | 429 |
| MAG | COM | 1 | 1-mag-post | 4/6/2011 | 8.00 | POST | 2294 | 150.5 | 534 |
| MAG | COM | 2 | 2-mag-pre | 5/2/2011 | 10.30 | PRE | 1281 | 84.1 | 298 |
| MAG | COM | 2 | 2-mag-post | 5/4/2011 | 11.00 | POST | 1393 | 91.4 | 324 |
| MAG | COM | 3 | 3-mag-pre | 5/23/2011 | 10.15 | PRE | 2087 | 136.9 | 486 |
| MAG | COM | 3 | 3-mag-post | 5/25/2011 | 10.15 | POST | 1142 | 74.9 | 266 |
| WIN | IND | 1 | 1-win-pre | 3/9/2011 | 8.19 | PRE | 1130 | 74.1 | 263 |
| WIN | IND | 1 | 1-win-post | 3/10/2011 | 14.00 | POST | 987 | 64.8 | 230 |
| WIN | IND | 2 | 2-win-pre | 4/6/2011 | 10.45 | PRE | 1000 | 65.6 | 233 |
| WIN | IND | 2 | 2-win-post | 4/7/2011 | 13.15 | POST | 963 | 63.2 | 224 |
| WIN | IND | 3 | 3-win-pre | 5/18/2011 | 15.00 | PRE | 937 | 61.5 | 218 |
| WIN | IND | 3 | 3-win-post | 5/19/2011 | 13.30 | POST | 760 | 49.9 | 177 |
| CLARK | SFR | 1 | 1-clark-pre | 4/4/2011 | 13.30 | PRE | 933 | 61.2 | 217 |
| JEFF | SFR | 1 | 1-jeff-pre | 3/8/2011 | 13.45 | PRE | 1112 | 73.0 | 259 |
| CLARK | SFR | 1 | 1-clark-post | 4/6/2011 | 9.30 | POST | 884 | 58.0 | 206 |
| JEFF | SFR | 1 | 1-jeff-post | 3/10/2011 | 10.10 | POST | 823 | 54.0 | 192 |
| CLARK | SFR | 2 | 2-clark-pre | 5/2/2011 | 13.15 | PRE | 1935 | 127.0 | 451 |
| JEFF | SFR | 2 | 2-jeff-pre | 4/5/2011 | 13.30 | PRE | 1246 | 81.8 | 290 |

| Site ID | Land Use | Event | Sample ID | Date | Time | Sweeping Type | Total Dry Accumulation | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Mass (g) | Loading (g/curb m) | Loading (lbs/curb mi) |
| CLARK | SFR | 2 | 2-clark-post | 5/4/2011 | 13.30 | POST | 1259 | 82.6 | 293 |
| JEFF | SFR | 2 | 2-jeff-post | 4/7/2011 | 10.30 | POST | 722 | 47.4 | 168 |
| CLARK | SFR | 3 | 3-clark-pre | 5/23/2011 | 11.15 | PRE | 1121 | 73.6 | 261 |
| JEFF | SFR | 3 | 3-jeff-pre | 5/24/2011 | 10.15 | PRE | 1085 | 71.2 | 253 |
| CLARK | SFR | 3 | 3-clark-post | 5/25/2011 | 12.30 | POST | 1143 | 75.0 | 266 |
| JEFF | SFR | 3 | 3-jeff-post | 5/26/2011 | 14.45 | POST | 1014 | 66.5 | 236 |

# Appendix B – City of Glendale Phase I Pre-Sweeping and Post-Sweeping Data

| Site ID | Land Use | Event | Sample ID | Date | Time | Pre or Post | Total Dry Accumulation | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Mass (g) | Loading (g/curb m) | Loading (lbs/curb mi) |
| VERD | ART | 1 | 1-verd-pre | 3/8/2011 | 9.30 | PRE | 1757 | 164.7 | 584 |
| VERD | ART | 1 | 1-verd-post | 3/10/2011 | 8.00 | POST | 1442 | 135.1 | 479 |
| VERD | ART | 2 | 2-verd-pre | 4/5/2011 | 8.15 | PRE | 1027 | 67.4 | 239 |
| VERD | ART | 2 | 2-verd-post | 4/7/2011 | 8.10 | POST | 1534 | 100.7 | 357 |
| VERD | ART | 3 | 3-verd-pre | 6/28/2011 | 9.45 | PRE | 1595 | 104.7 | 371 |
| VERD | ART | 3 | 3-verd-post | 6/30/2011 | 8.10 | POST | 2326 | 152.6 | 541 |
| ORNGE | COM | 1 | 1-ornge-pre | 4/12/2011 | 7.45 | PRE | 398 | 26.1 | 93 |
| ORNGE | COM | 1 | 1-ornge-post | 4/14/2011 | 7.45 | POST | 211 | 13.8 | 49 |
| ORNGE | COM | 2 | 2-ornge-pre | 5/3/2011 | 10.30 | PRE | 418 | 27.4 | 97 |
| ORNGE | COM | 2 | 2-ornge-post | 5/5/2011 | 9.00 | POST | 485 | 31.8 | 113 |
| ORNGE | COM | 3 | 3-ornge-pre | 5/10/2011 | 8.00 | PRE | 250 | 16.4 | 58 |
| ORNGE | COM | 3 | 3-ornge-post | 5/12/2011 | 8.00 | POST | 443 | 29.1 | 103 |
| GRAND | IND | 1 | 1-grand-pre | 2/28/2011 | 7.00 | PRE | 811 | 53.2 | 189 |
| GRAND | IND | 1 | 1-grand-post | 3/2/2011 | 7.35 | POST | 827 | 54.3 | 193 |
| GRAND | IND | 2 | 2-grand-pre | 4/11/2011 | 8.00 | PRE | 1865 | 122.4 | 434 |
| GRAND | IND | 2 | 2-grand-post | 4/13/2011 | 7.45 | POST | 874 | 57.3 | 203 |
| GRAND | IND | 3 | 3-grand-pre | 5/9/2011 | 8.00 | PRE | 1117 | 73.3 | 260 |
| GRAND | IND | 3 | 3-grand-post | 5/11/2011 | 8.00 | POST | 564 | 37.0 | 131 |
| KIRK | SFR | 1 | 1-kirk-pre | 2/28/2011 | 8.40 | PRE | 462 | 30.3 | 108 |
| KIRK | SFR | 1 | 1-kirk-post | 3/2/2011 | 9.15 | POST | 520 | 34.1 | 121 |
| KENN | SFR | 1 | 1-kenn-pre | 4/4/2011 | 8.15 | PRE | 1271 | 83.4 | 296 |
| KENN | SFR | 1 | 1-kenn-post | 4/5/2011 | 10.30 | POST | 1742 | 114.3 | 406 |
| KIRK | SFR | 2 | 2-kirk-pre | 4/11/2011 | 10.00 | PRE | 559 | 36.7 | 130 |
| KIRK | SFR | 2 | 2-kirk-post | 4/13/2011 | 9.30 | POST | 1068 | 70.1 | 249 |

| Site ID | Land Use | Event | Sample ID | Date | Time | Pre or Post | Total Dry Accumulation | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Mass (g) | Loading (g/curb m) | Loading (lbs/curb mi) |
| KENN | SFR | 2 | 2-kenn-pre | 5/2/2011 | 8.30 | PRE | 1911 | 125.4 | 445 |
| KENN | SFR | 2 | 2-kenn-post | 5/3/2011 | 9.15 | POST | 3172 | 208.1 | 738 |
| KIRK | SFR | 3 | 3-kirk-pre | 5/9/2011 | 9.30 | PRE | 556 | 36.5 | 130 |
| KIRK | SFR | 3 | 3-kirk-post | 5/11/2011 | 9.30 | POST | 1117 | 73.3 | 260 |
| KENN | SFR | 3 | 3-kenn-pre | 6/27/2011 | 8.25 | PRE | 2515 | 165.0 | 585 |
| KENN | SFR | 3 | 3-kenn-post | 6/28/2011 | 8.10 | POST | 685 | 44.9 | 159 |

# Appendix C – Phase II Particle Size Distribution Data

| | | | | | | | Mass Fraction (%) in Particle Size Group | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fine | Medium | | | Coarse | | | Not in Group |
| | | | | | | | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| City | Sweeper | Target Speed (mph) | Actual Speed (mph) | Initial Loading (lbs/curb mi) | Final Loading (lbs/curb mi) | Removal (%) | <63µm | 63-125µm | 125-250µm | 250-600µm | 600-1000µm | 1000-2000µm | 2000-6300µm | >6300µm |
| Burbank | Schwarze M6000 | 5 mph | 4.55 | 634 | 175 | 72 | 8.74 | 13.82 | 32.67 | 24.37 | 12.26 | 6.03 | 2.11 | 0.00 |
| Burbank | Schwarze M6000 | 5 mph | 5.41 | 317 | 229 | 28 | 14.79 | 17.34 | 37.57 | 18.68 | 7.48 | 3.27 | 0.87 | 0.00 |
| Burbank | Schwarze M6000 | 10 mph | 9.58 | 634 | 225 | 65 | 9.87 | 14.83 | 30.50 | 21.08 | 12.95 | 8.04 | 2.73 | 0.00 |
| Burbank | Schwarze M6000 | 10 mph | 10.49 | 317 | 136 | 57 | 31.23 | 21.72 | 31.36 | 10.05 | 3.26 | 1.77 | 0.61 | 0.00 |
| Burbank | Schwarze A7000 | 5 mph | 5.3 | 634 | 188 | 70 | 18.67 | 10.93 | 20.13 | 14.00 | 12.96 | 12.33 | 10.98 | 0.00 |
| Burbank | Schwarze A7000 | 5 mph | 4.08 | 317 | 56 | 82 | 26.56 | 14.26 | 21.73 | 15.16 | 10.33 | 7.47 | 4.49 | 0.00 |
| Burbank | Schwarze A7000 | 10 mph | 10.79 | 634 | 118 | 81 | 28.89 | 14.27 | 26.01 | 15.53 | 8.07 | 4.94 | 2.29 | 0.00 |
| Glendale | Elgin Crosswind | 5 mph | 7.07 | 634 | 155 | 76 | 27.03 | 17.82 | 28.62 | 13.29 | 7.50 | 4.25 | 1.49 | 0.00 |
| Glendale | Elgin Crosswind | 5 mph | 7.12 | 634 | 101 | 84 | 20.54 | 11.64 | 21.91 | 18.67 | 14.61 | 8.79 | 3.84 | 0.00 |
| Glendale | Elgin Crosswind | 5 mph | 6.74 | 317 | 69 | 78 | 23.86 | 19.05 | 35.89 | 13.28 | 5.03 | 2.32 | 0.57 | 0.00 |
| Glendale | Elgin Crosswind | 10 mph | 10.39 | 634 | 132 | 79 | 23.71 | 19.10 | 36.64 | 14.31 | 4.56 | 1.62 | 0.06 | 0.00 |
| Glendale | Elgin Crosswind | 10 mph | 11.18 | 317 | 62 | 80 | 24.86 | 16.93 | 33.81 | 14.93 | 6.14 | 2.87 | 0.46 | 0.00 |
| Glendale | Allianz CNG 4000 | 5 mph | 5.4 | 634 | 194 | 69 | 21.32 | 18.75 | 37.46 | 15.19 | 4.28 | 2.06 | 0.83 | 0.11 |
| Glendale | Allianz CNG 4000 | 5 mph | 5.29 | 317 | 95 | 70 | 28.92 | 17.25 | 28.26 | 13.70 | 6.82 | 3.74 | 1.31 | 0.00 |
| Glendale | Allianz CNG 4000 | 10 mph | 10.79 | 634 | 133 | 79 | 28.57 | 19.15 | 27.01 | 12.85 | 5.93 | 4.24 | 2.25 | 0.00 |
| Glendale | Elgin Broome Bear | 5 mph | 4.96 | 634 | 160 | 75 | 16.08 | 22.17 | 38.64 | 15.70 | 4.99 | 2.09 | 0.33 | 0.00 |
| Glendale | Elgin Broome Bear | 10 mph | 9.66 | 634 | 184 | 71 | 17.18 | 18.52 | 39.89 | 16.49 | 5.57 | 2.19 | 0.16 | 0.00 |
| Glendale | Elgin Broome Bear | 10 mph | 10 | 317 | 126 | 60 | 17.97 | 19.70 | 33.43 | 18.34 | 6.64 | 3.29 | 0.63 | 0.00 |

# Appendix D – Phase III Particle Size Distribution Data

| | | | | | Mass Fraction (%) in Particle Size Group | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Fine | | Medium | | Coarse | | Not in Group | |
| | | | | | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| City | Site ID | Land Use | Pre or Post | Sweeper | <63µm | 63-125µm | 125-250µm | 250-600µm | 600-1000µm | 1000-2000µm | 2000-6300µm | >6300µm |
| Burbank | JEFF | SFR | PRE | Schwarze A7000 / 10 mph | 13.67 | 16.14 | 22.49 | 24.05 | 10.38 | 7.50 | 4.04 | 1.73 |
| Burbank | JEFF | SFR | POST | Schwarze A7000 / 10 mph | 14.82 | 16.82 | 22.66 | 22.81 | 9.52 | 6.22 | 5.61 | 1.54 |
| Burbank | WIN | IND | PRE | Schwarze A7000 / 10 mph | 13.38 | 16.97 | 21.70 | 23.67 | 10.77 | 8.66 | 4.08 | 0.77 |
| Burbank | WIN | IND | POST | Schwarze A7000 / 10 mph | 13.38 | 14.92 | 17.90 | 22.23 | 12.01 | 12.37 | 6.30 | 0.89 |
| Glendale | KENN | SFR | PRE | Elgin Crosswind / 10 mph | 7.85 | 12.83 | 21.55 | 29.49 | 12.13 | 10.12 | 5.24 | 0.79 |
| Glendale | KENN | SFR | POST | Elgin Crosswind / 10 mph | 12.93 | 17.14 | 23.34 | 24.55 | 9.87 | 7.65 | 3.60 | 0.92 |